

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Computer Applications (BCA)

Semester-V University Examination November-2019

CA 314 Object Oriented Programming Through JAVA

Date: 21.11.19, Thursday Time: 10.00 a.m. to 01.00 p.m. Maximum Marks: 70

Instructions:

1. Attempt both sections in separate answer books.
2. Figure to right indicates marks.
3. Make suitable assumption when necessary.

SECTION – 1

Q.1 (A) Do as directed(Each carries One mark) [05]

1. Which of these can be used to fully abstract a class from its implementation?
a). Objects b). Packages
c). Interfaces d). None of these.
2. Which component is responsible to optimize bytecode to machine code?
a). JVM b). JDK
c). JIT d). JRE
3. Which of these is supported by method overriding in Java?
a). Abstraction b). Encapsulation
c). Polymorphism d). None of the mentioned
4. Which of these class is used to create object whose character sequence is mutable?
a). String() b). StringBuffer()
c). Both a & b. d). None of these
5. Can a final variable be added?
final a = 10; final b = 20; System.out.println(a+b);
a). Error. b). 30.
c). 10 20. d). 0.

(B) Define the following: [06]

- a). JVM.
- b). Garbage Collection.

Q.2 (A) What is method overloading ? Explain with suitable example. [04]

(B) Write a program to count number of object created for a class using initializer block. [04]

(C) How multiple inheritance is implemented in Java? Explain with suitable code. [04]

OR

Q.2 (A) What is Constructor? How the constructor gets executed in multilevel inheritance. [04]

(B) What is the use of 'this' keyword ? Explain with example. [04]

(C) Differentiate between abstract class and interface. [04]

Q.3 (A) Write a program to accept a string from user , reverse the string and check whether the input string is Palindrome or not. [06]

(B) What is DMD? Explain with suitable example using abstract class. [06]

OR

Q.3 (A) Write a program to find the area of Circle, Rectangle, Square using Runtime Polymorphism using interface. [06]

(B) Explain the use of super keyword with suitable example. [06]

SECTION – 2

Q.4 (A) Do as directed (Each Carries One mark)

[05]

1. Which of these class is related to all the exceptions that cannot be caught?
 - a). Error
 - b). Exception
 - c). RuntimeExecption
 - d). All of the mentioned
2. Which of the following will directly stop the execution of a Thread?
 - a). wait()
 - b). notify()
 - c). notifyAll()
 - d). exits synchronized code
3. Converting a primitive type data into its corresponding wrapper class object
 - a). Boxing
 - b). Wrapping
 - c). Instantiation
 - d). Autoboxing
4. Program which executes applet is known as
 - a). Applet engine
 - b). Virtual machine
 - c). JVM
 - d). None of above
5. Which of the following are passed as an argument to the paint() method?
 - a). A Canvas object.
 - b). A Graphics object.
 - c). An Image object.
 - d). A Paint object

(B) Define the following:

b). Exception.

[06]

Q.5 (A) What is synchronization in java? Explain with suitable example.

[04]

(B) What is Package? State the advantages of Packages.

[04]

(C) Explain Life cycle of a Thread with suitable diagram.

[04]

OR

Q.5 (A) What do you mean by Checked & Unchecked Exceptions?

[04]

(B) What are Thread priorities? What is defined by the priority of Thread?

[04]

(C) What is applet? Explain the advantage of applet.

[04]

Q.6 (A) Write a java code to demonstrate Moving Banner using suitable example.

[06]

(B) Explain the five keywords used in exception handling.

[06]

OR

Q.6 (A) Write a class Driver with attributes vehicle no, name & age. Initialize values through parameterized constructor. If age of driver is less than 18 then generate user-defined exception “AgeNotWithinTheRange”.

[06]

(B) Explain Life cycle of an applet .

[06]